



PhenoRx

Pharmacogenomic Adverse Drug Reaction Risk Assessment

A technical and methodological overview of a clinical decision support prototype that combines a machine-learning classifier, pharmacogenomic rule engines, and a large language model to estimate adverse drug reaction risk before a medicine is administered.

Written for medical students and clinicians with no machine-learning background. It explains what the system does, how each component works, and — importantly — how well it actually performs.

This is a teaching and research prototype, not a clinical tool.

It is trained on synthetic data, has never been externally validated, and its measured accuracy is below that of a trivial baseline. It must not be used to make decisions about any real patient. Section 6 explains the evidence for this statement in full.

Document	Technical & Methodological Overview
System version	2.1.0
Model artifact	<code>models/adr_model.pkl</code> (2.5 MB)
Audience	Medical students, clinical pharmacology trainees

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How to read this document

Sections 1–3 describe the system and need no technical background. Sections 4–7 go into the methods. **Section 6 is the most important section** — it reports the measured performance, separates the two different questions the model is asked, and derives the mathematical ceiling on how well either can be answered. If you read only one part, read that one: it is the part that teaches you how to interrogate any clinical AI tool you are shown in future.

1 The clinical problem

An adverse drug reaction (ADR) is harm caused by a medicine given at a normal dose for a normal indication. ADRs are a substantial cause of hospital admission and inpatient morbidity, and a meaningful proportion are considered preventable.

The clinical difficulty is that risk is not uniform. Two patients given the same drug at the same dose can have very different outcomes, because risk depends on the interaction of:

- **Organ function** — chiefly renal clearance (eGFR) and hepatic metabolism, which determine how quickly a drug is eliminated.
- **Genetics** — inherited variation in metabolising enzymes and immune receptors, which can multiply exposure or predispose to severe immune reactions.
- **Co-medication** — competing substrates, enzyme inhibition or induction, and additive toxicity across a regimen.
- **Comorbidity and physiology** — age, body composition, cardiac disease, electrolyte state.

A clinician integrates these mentally. The premise of this project is that the same integration can be performed computationally, before the first dose, to flag patients who warrant a different agent, a lower dose, or closer monitoring.

The premise is reasonable; the execution here is a prototype

The idea above is sound and is an active area of real research. This particular implementation does not achieve it — see Section 6. The value of this document is in showing how such a system is constructed and how to evaluate whether it works.

Why "phenotype" plus "Rx"

The name reflects the intended approach: combine the patient's observable clinical phenotype (laboratory values, comorbidities, physiology) with their pharmacogenomic phenotype (metaboliser status) to individualise the prescription.

2 What the system does

The application is a web tool organised around a single clinical assessment. A user works through five screens in sequence.

1 • Patient details

Demographics: age, sex, ethnicity, height and weight. Body mass index is calculated automatically, and age and sex select the laboratory reference intervals used on every subsequent screen.

2 • Clinical assessment

The main data-entry form: renal and hepatic panels, full blood count and differential, vital signs, comorbidities, the drug regimen, and pharmacogenomic results. Values are validated against the same numeric bounds the server enforces, so an implausible entry is flagged as it is typed.

~60 fields, fillable in any order; a demo patient can populate all of them

3 • Report

The model's numerical output and the AI-generated narrative are presented side by side: overall risk percentage, risk tier, predicted reaction classes, contributing factors, and pharmacogenomic profile. Exportable as PDF.

4 • Interactions and dosing

Pairwise interaction checking across the regimen with severity classification, plus renal-, hepatic- and CYP-adjusted dose suggestions with monitoring intervals.

5 • Counselling

A six-step structured medication education workflow, including teach-back comprehension scoring and signed acknowledgement from both patient and counsellor.

Risk tiers

The model returns a probability, which is mapped to one of four tiers. The thresholds are fixed constants in the code, not learned from data:

Risk tier thresholds, from `services/ml_service.py`

TIER	OVERALL ADR RISK	INTENDED INTERPRETATION
Critical	≥ 75%	Consider an alternative agent; specialist input
High	≥ 50%	Dose reduction and active monitoring
Moderate	≥ 25%	Standard dosing with scheduled review
Low	< 25%	Routine care

A point worth noticing

These cut-points are arbitrary round numbers. In a validated clinical tool, thresholds would be derived from the data — chosen to achieve a target sensitivity or to optimise net clinical benefit, then tested on a separate cohort. Fixed round numbers are a reasonable placeholder in a prototype, but they are an assumption, not a finding.

3 Architecture

Three independent layers contribute to the output. They are worth separating clearly, because they have very different reliability characteristics.

LAYER	WHAT IT IS	HOW IT CAN FAIL
1. ML classifier	A gradient-boosting model trained on 50,000 synthetic patient records	Silently wrong. Produces a confident-looking number whether or not it is correct, and cannot signal its own uncertainty. See Section 6 for how well it performs.
2. PGx rule engines	Deterministic rules encoding published gene–drug relationships	Transparent. Rules are explicit and auditable; failure means a rule is missing or out of date, which is inspectable.
3. LLM narrative	A large language model that writes the clinical prose	Fluent but unreliable. Can produce plausible, well-written statements that are incorrect. Degrades to fixed rule-based text when unavailable.

Technology stack

Backend	Python 3.13, Flask (application factory with blueprints)
Machine learning	scikit-learn — <code>HistGradientBoostingClassifier</code>
Frontend	React 19 with Vite, Tailwind CSS, single-page application
Language model	NVIDIA NIM (Nemotron), OpenAI-compatible API
Serving	Gunicorn, one worker with four threads, containerised
Persistence	In-memory only — no database

Why the three layers are kept separate

Separation means a failure in one layer does not silently corrupt the others. If the language model is unreachable, the system still returns the model's number and the pharmacogenomic findings, with deterministic text in place of the narrative. The converse also holds: a fluent AI narrative does not make the underlying risk number any more trustworthy.

4 Layer 1 — The machine-learning model

The algorithm

The model is a **histogram-based gradient boosting classifier** (`HistGradientBoostingClassifier` from scikit-learn). In clinical terms, it can be understood as follows.

A single *decision tree* is a flowchart of yes/no questions — "is eGFR below 30?", "is the patient over 70?" — ending in a prediction. One tree alone is crude. *Gradient boosting* builds hundreds of trees in sequence, where each new tree is trained specifically on the cases the existing trees got wrong. The final prediction is the accumulated vote of all the trees.

This family of model is a sensible default for tabular clinical data: it handles mixed numeric and categorical variables, tolerates missing values natively, and captures interactions between variables without being told to look for them.

Training hyperparameters, from `config/ml_config.yaml` defaults

PARAMETER	VALUE	MEANING
<code>max_iter</code>	400	Maximum number of boosting trees
<code>max_depth</code>	18	Maximum questions deep any one tree may go
<code>learning_rate</code>	0.07	How much each tree is allowed to correct its predecessors
<code>l2_regularization</code>	0.3	Penalty discouraging over-confident fitting
<code>random_state</code>	42	Fixed seed, so training is reproducible
train/test split	80 / 20	A fifth of data held back to test on

Input features

The model consumes **29 variables** per patient:

DOMAIN	VARIABLES
Demographics	age, sex, ethnicity, weight, height, BMI
Renal	creatinine, eGFR
Hepatic	AST/ALT, bilirubin, albumin
Haematology	haemoglobin, white cell count, platelet count
Vital signs	systolic and diastolic blood pressure, heart rate, temperature
Comorbidity	diabetes, liver disease, chronic kidney disease, cardiac disease, hypertension
Drug exposure	index drug dose, concomitant drug count, polypharmacy flag, days since start
Pharmacogenomic	CYP2C9 and CYP2D6 metaboliser status

Note the mismatch between form and model

The assessment form collects roughly 60 fields, including an extensive pharmacogenomic panel (six CYP enzymes, three transporters, three HLA alleles). The model itself uses only the 29 variables above, of which just **two** are pharmacogenomic. The remaining PGx data is used by the rule engines in Layer 2, not by the model. This is worth knowing: entering a full genomic panel does not make the risk *number* more informed than entering CYP2C9 and CYP2D6 alone.

Training data

Records	50,000
Origin	Synthetic — generated by <code>scripts/data_generator.py</code>
Outcome classes	20 distinct ADR types, including "No ADR"
Split	40,000 training / 10,000 held-out test
Reaction rate	31.9% of records carry a reaction
Risk assignment	Continuous, from a log-odds sum over clinical factors

The size of the training set was itself the main defect

Until recently the model was trained on a **200-row** file spanning **60** outcome classes — roughly three examples per class. The data generator defaults to 200,000 records; it had been run with 200. Regenerating the cohort at 50,000 records reduced the class count to 20, raised the rarest class from 3 examples to 265, and is the single change responsible for most of the improvement reported in Section 6.

The data is synthetic, and this is the root limitation

No real patient contributed to this model. The records were produced by a script that encodes assumptions about how clinical variables relate to adverse outcomes. A model trained this way can only learn the rules its generator already contained — it cannot discover anything about real physiology, and good performance on such data would demonstrate nothing about real-world accuracy.

5 Layer 2 — Pharmacogenomic rule engines

Separate from the model, four deterministic engines encode published gene–drug relationships. These are ordinary conditional logic, not learned from data, which makes them transparent and auditable.

ENGINE	SCOPE
CYP engine	Cytochrome P450 metaboliser phenotypes. Poor metabolisers accumulate an active drug; ultra-rapid metabolisers may fail to respond, or over-produce an active metabolite from a prodrug.
Transporter engine	Membrane transporter variants affecting drug distribution and hepatic uptake — the mechanism underlying, for example, statin-associated muscle symptoms.
HLA engine	Human leukocyte antigen alleles associated with severe immune-mediated hypersensitivity, the class of reaction where pre-prescription genotyping has the clearest established clinical value.
Non-CYP engine	Other metabolising enzymes outside the P450 system.

Why this layer is more trustworthy than the model

A rule stating a relationship between a genotype and a drug can be traced to a published source, reviewed by a pharmacologist, and corrected when guidance changes. Its behaviour is identical every time and can be reasoned about without running it. The machine-learning model has none of these properties: it is a numerical function fitted to data, and its reasoning cannot be inspected directly.

The generalisable lesson

When you encounter a clinical AI product, ask which parts are learned and which are encoded rules. Rules can be checked against guidelines. Learned components can only be trusted to the extent that they have been validated on data resembling your patients.

6 Model performance — the honest numbers

This section reports what the model actually scores on a held-out fifth of the data it never saw in training. Two questions are measured separately, because the system answers both and they are not equally difficult.

6.1 What the model scores

Held-out test set, 10,000 records. Measured from the shipped artifact.

QUESTION	RESULT	BASELINE	CEILING	FIELD
Will this patient react? Binary — accuracy	80.38%	67.89%	80.85%	<code>overall_adr_risk</code>
How well does it rank risk? Binary — ROC-AUC	0.834	0.500	0.840	<code>overall_adr_risk</code>
Which reaction will it be? 20-class — accuracy	70.92%	67.89%	—	<code>predicted_adr_type</code>

The **baseline** is a trivial rule that ignores every input and always answers the commonest class. The **ceiling** is the best any model could achieve, and is calculable here because the data was made by a known process (§6.3).

The model has captured almost all the available signal

80.38% against a ceiling of 80.85%, and ROC-AUC 0.834 against a ceiling of 0.840. Roughly 99% of what is learnable has been learned. Further tuning of the algorithm would achieve essentially nothing; the remaining gap is not a modelling deficiency.

6.2 The headline number

80.4%

ACCURACY
Baseline 67.9%

0.834

ROC-AUC
0.5 = coin toss

78.5%

PRECISION
Flags that were right

53.6%

RECALL
Reactions caught

An ROC-AUC of 0.834 means that given one patient who reacted and one who did not, the model assigns the higher risk to the right one about 83% of the time. That is respectable discrimination.

Recall of 53.6% is the number a clinician should care about

At the default 50% threshold the model misses nearly half of the reactions that occur. For a safety tool that is the wrong balance: missing a reaction is worse than an unnecessary warning. Lowering the threshold trades precision for recall, and choosing that operating point is a clinical decision, not a technical one — which is exactly what §6.5 says is still missing.

6.3 Why a ceiling exists

Each synthetic patient is assigned a reaction *probability* from their clinical picture — organ function, comorbidity, genotype, regimen — and the outcome is then drawn against it. A patient at 30% risk reacts in three runs out of ten.

This means the outcome is genuinely uncertain even with perfect knowledge. A model that had recovered the risk function exactly would still be wrong whenever the draw went against the more likely outcome. Averaging that irreducible error across the cohort gives the **80.85% ceiling**.

Why this matters beyond this project

Every clinical prediction has such a ceiling, because outcomes are not fully determined by what can be measured. Here it is calculable because the generating process is known; with real patients it is not, but it still exists. A tool claiming 99% accuracy on a genuinely uncertain outcome is telling you its evaluation is flawed, not that it is exceptional.

6.4 How the risk model was arrived at

Getting to these figures took three attempts at the data generator, and the failures are more instructive than the result.

Three generations of the synthetic risk model

DESIGN	BEHAVIOUR	ACCURACY	AUC
Deterministic branches	Matching a condition guaranteed a reaction, so the model learned $P(\text{reaction} \mid \text{CKD}) = 1.0$ and reported ~100% risk for any patient with any comorbidity . Scored well; useless at the bedside.	85.4%	0.930
Fixed incidence per branch	Clinically plausible rates, but only seven distinct risk values existed across the cohort — every CKD patient identical regardless of eGFR, age or regimen. Nothing to learn.	73.2%	0.609
Continuous risk	Risk is a log-odds sum over individual factors, so it rises smoothly with severity and two patients sharing a diagnosis differ by everything else about them.	80.4%	0.834

The highest-scoring version was the least usable

The deterministic design scored 85.4% with an AUC of 0.930 — the best numbers of the three — while telling a clinician that every patient with a comorbidity was at ~100% risk. Its scores were high precisely *because* its rules were trivially learnable, and a rule stating that chronic kidney disease guarantees an adverse reaction is clinically false.

This is the clearest lesson in this document: **a headline metric can improve while the tool becomes less useful**. Ask what a model outputs for a realistic patient, not only what it scores.

6.5 Do the risk scores behave sensibly?

A metric cannot show whether output is clinically coherent. The table below takes one synthetic patient and varies a single factor at a time:

Predicted `overall_adr_risk`, one factor changed at a time

PATIENT	RISK	TIER	PREDICTED REACTION
Healthy adult, 27, ibuprofen	4.6%	Low	Gastrointestinal
+ six concomitant drugs	6.5%	Low	Gastrointestinal
+ aged 82	8.0%	Low	Gastrointestinal
+ cardiac disease	8.0%	Low	Gastrointestinal
+ eGFR 55	9.1%	Low	Gastrointestinal
+ CKD, eGFR 28	21.7%	Low	Nephrotoxicity
Elderly, CKD eGFR 28, eight drugs	53.4%	High	Nephrotoxicity

Risk accumulates rather than saturating, and the predicted reaction shifts to nephrotoxicity exactly when renal function becomes the dominant problem. Across the test set the scores spread properly: 5% at the 5th percentile, 18% at the median, 94% at the 95th.

The risk tiers no longer match the distribution

Tier cut-points are fixed at 25 / 50 / 75%, but the median patient now scores 18%, so most land in "Low". The thresholds were chosen as round numbers against a different risk distribution and have not been re-derived. They should be set from a target sensitivity on real data.

6.6 Claims that were removed

Two fabricated metrics used to be written into every report

`scripts/model_trainer.py` emitted these as fixed text regardless of measured results — "> 92% recall" for pharmacogenomic hypersensitivities, and an ablation reporting "weighted F1 0.871 → 0.774". Neither was supported by any artifact in the repository. Both are deleted, and the report generator now emits only figures produced by the run itself.

Recognising this pattern is the transferable skill: a stated metric is only as good as the experiment behind it, and a number that cannot be traced to a reproducible run should not be believed.

6.7 What is still missing

Every figure above is an honest measurement on **synthetic** data. They describe how well the model recovers a risk function that was written by hand, and say nothing about real patients. Clinical credibility would still require:

1. **Real data** with linked outcomes, under appropriate governance.
2. **Calibration assessment** — a stated 70% risk should correspond to the event occurring in about 70% of such patients.
3. **A justified operating threshold** — 53.6% recall is too low for a safety tool; the cut-point should come from a target sensitivity.
4. **Re-derived risk tiers**, matched to the real risk distribution.
5. **External validation** on a different institution's population.

6. **Prospective evaluation** of whether using the tool changes outcomes.

7 Layer 3 — The AI narrative

The third layer uses a large language model to convert the numerical output into clinical prose: an interpretation, monitoring suggestions, and a mitigation plan.

Provider	NVIDIA NIM (OpenAI-compatible endpoint)
Primary model	nvidia/nemotron-3-super-120b-a12b
Fallback model	nvidia/nemotron-3-ultra-550b-a55b
Final fallback	Deterministic rule-based text

Graceful degradation

Language model endpoints fail routinely — they return "temporarily overloaded", or take too long. The system tries the primary model, retries once if it reports itself overloaded, falls through to the secondary model, and finally emits fixed rule-based text. A total time budget bounds the whole sequence so a request cannot hang.

Every response carries an `ai_generated` flag, so it is always possible to tell whether the text was written by the model or by the deterministic fallback.

Fluency is not accuracy

A language model produces text that reads as though written by a confident clinician, whether or not the content is correct. It has no access to the patient beyond the fields supplied, and it cannot verify the risk score it is asked to explain. The risk score it is given now carries real signal (Section 6), but it is derived from synthetic training data, and the narrative will read with exactly the same confidence whether the underlying number is sound or not. Every AI-generated statement requires independent verification.

8 Safety, limitations and governance

Not suitable for clinical use

This system must not inform decisions about any real patient. The limitations below are not caveats around a working tool; several are fundamental.

Model limitations

- **Trained entirely on synthetic data** — no real patient contributed, so no figure in Section 6 describes real-world performance. This remains the fundamental limitation, and no amount of tuning addresses it.
- **Never externally validated** — no prospective or independent evaluation.
- **Uncalibrated** — the stated percentages have no verified relationship to real event frequencies.
- **Recall is only 53.6%** at the default threshold — nearly half of reactions are missed. For a safety tool that is the wrong direction of error, and the operating point has not been chosen against any clinical target.
- **Risk tiers are arbitrary** — 25/50/75% are round numbers, not thresholds derived to achieve a target sensitivity.

System limitations

- **No authentication** — every endpoint is publicly accessible.
- **No database** — assessments are held in memory and lost on restart.
- **No audit trail** — a clinical requirement that is entirely absent.
- **No regulatory clearance** — a tool of this type would be a regulated medical device in most jurisdictions.

What the project does demonstrate

It is a working example of a full clinical decision support architecture: validated data capture, a served ML model, deterministic rule engines, language model integration with sensible failure handling, and a documented deployment. The model now discriminates genuinely on the task it was given, and Section 6.4 is itself instructive — the largest single gain came not from a better algorithm but from noticing the training set had 200 rows instead of 200,000. As a teaching artifact for how such systems are built, and how to evaluate the claims they make, it is useful. As a clinical instrument, it is not.

What to take away

- Always ask what baseline an accuracy figure beats.
- Ask whether the training data is real, and whether validation was external.
- For rare outcomes, demand per-class recall, not overall accuracy.
- Check whether a quoted metric can be traced to a reproducible run.
- Treat fluent AI prose as a presentation layer, never as evidence.

9 Glossary

TERM	IN CLINICAL LANGUAGE
Accuracy	Proportion of all predictions that were correct. Misleading when one outcome dominates.
Precision	Positive predictive value — of those flagged, how many truly had the outcome.
Recall / sensitivity	Of those who truly had the outcome, how many were detected.
F1 score	Harmonic mean of precision and recall; a single number balancing the two.
Support	Number of true cases of a class in the test set — the denominator behind the percentages.
Baseline	The score achievable by a trivial rule. A model must beat it to be worth anything.
Class imbalance	One outcome vastly outnumbering others — as with any rare disease.
Calibration	Whether a stated 70% risk corresponds to the event occurring in 70% of such patients.
Training / test split	Holding data back to test on, so performance is not measured on cases already seen.
External validation	Testing on a different population or institution — the standard for real clinical claims.
ROC-AUC	Given one patient with the outcome and one without, the chance the model ranks them correctly. 0.5 is a coin toss, 1.0 is perfect.
Balanced accuracy	Mean recall across all classes, so a rare class counts as much as a common one.
Bayes-optimal ceiling	The best score any model could achieve, set by irreducible randomness in the outcome.
Class weighting	Telling the model to treat rare classes as equally important, trading overall accuracy for rare-class detection.
Gradient boosting	Many simple decision trees in sequence, each correcting its predecessors' errors.
Feature	One input variable, e.g. eGFR or age.
Hyperparameter	A setting chosen before training, e.g. how many trees to build.
Synthetic data	Computer-generated records, not from real patients.
Large language model	A text-generating model. Fluent by construction; not a source of verified fact.
Pharmacogenomics	How inherited genetic variation alters drug response.
CYP enzymes	Cytochrome P450 — the main family metabolising most drugs.
Poor metaboliser	Reduced enzyme activity, so the drug accumulates at standard doses.
HLA allele	Immune system variant; specific alleles predict severe hypersensitivity to particular drugs.

PhenoRx Technical & Methodological Overview · System version 2.1.0. All performance figures in this document were taken from the repository's own artifacts or measured directly from the shipped model; none are estimated. The model evaluated is `models/adr_model.pkl`.

This document describes a research and teaching prototype. It is not a clinical tool, has no regulatory clearance, and must not be used in the care of any patient.